MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCFSC201 - FORENSICPHYSICS (2024 ADMISSION ONWARDS)

Duration: 2 hrs

Maximum Marks: 70

Part A – Short Answers

1. Tricalcium Silicate
2. Density Gradient Tube Method (or Soil Fractionation)
3. Bitumen (or Asphalt/Binder)
4. Speech Enhancement (or Noise Reduction/Filtering)
5. Speaker variability refers to natural changes in one's voice; disguise is an intentional attempt to alter voice identity.
6. Signal-to-Noise Ratio (SNR)
7. Voice Quality (or Fundamental Frequency/Pitch)
8. Point of Impact (POI)
9. Voice Comparison (or Speaker Identification/Verification)
10. Silica (Silicon Dioxide)

Part B: Short Answer Questions
Answer any 5. Each carries 3 marks.

11. Sealing and Labeling Evidence:

Place evidence in primary container (paper bindle/bag); seal with tamper-evident tape. (1.5 Marks) Label with case number, date, item description, and investigator's signature across the seal. (1.5 Marks)

12. Glass Fragments Movement:

Explanation of radial and concentric fractures. (1 Mark)

Elasticity causes glass to spring back; "backscatter" or "back spatter" occurs when fragments are ejected toward the force. (2 Marks)

13. Weathering in Soil Formation:

Physical weathering: Breakdown of rocks via temperature/pressure. (1 Mark)
Chemical weathering: Alteration of minerals via water/oxygen. (1 Mark)
Forensic significance: Creates unique mineral profiles. (1 Mark)

14. Forensic Examination of Electrical Installations:

Checking for arcing, short circuits, or overloaded circuits. (1.5 Marks)
Examination of fuses, circuit breakers, and "beading" on wires to determine fire origin. (1.5 Marks)

15. Speech Production Process:

Three stages: Respiration (lungs), Phonation (vocal folds), and Articulation (mouth/tongue). (3 Marks)

16. Transcription of Disputed Utterances:

Listening in controlled acoustic environments; use of phonetic transcription (IPA). (1.5 Marks)
Documenting context, background noise, and uncertain words in brackets. (1.5 Marks)

17. Admissibility of Voice Evidence:

Standards like Daubert or Frye (depending on jurisdiction); requirement for expert testimony. (1.5 Marks)
The "chain of custody" and proof that the recording hasn't been tampered with. (1.5 Marks)

Part C: Short Essay Questions

Answer any 5. Each carries 6 marks.

18. Interpretation of Soil Evidence: Comparison of color, texture, and mineralogy between crime scene and suspect samples.

19. Evidentiary Value of Glass: Discussion of refractive index (RI), density, and physical matching (jigsaw fit).

20. Glass Fracture Mechanism: Detailed explanation of stress/strain, 3R rule (Radial cracks, Right angles, Rear), and impact direction.

21. Feature Extraction: Focus on Mel-Frequency Cepstral Coefficients (MFCCs), pitch, and formants.

22. Pattern Recognition: Principles of template matching and statistical modeling (e.g., GMM-UBM) in speech.

23. Legal Aspects of Audio Evidence: Authentication, wiretapping laws, and the Indian Evidence Act (or relevant local law) regarding digital recordings.

24. Cement Concrete Failure: Analysis of mixing ratios, curing issues, and environmental degradation leading to collapse.

Part D: Essay Questions

Answer any 1. Carries 15 marks.

25. Soil Comparison Techniques:
    - **Particle Size:** Sieve analysis and sedimentation rates. (5 Marks)
    - **Turbidity:** Measuring suspended particles for chemical consistency. (5 Marks)
    - **Microscopy:** Identifying unique minerals, biological matter, and man-made debris. (5 Marks)
26. Basic Factors of Sound in Speech Acoustics:
    - **Frequency/Pitch:** Biological markers of the speaker. (5 Marks)

- **Intensity/Loudness:** Contextual and emotional indicators. (5 Marks)
- **Time/Duration:** Rhythmic patterns and speech rate analysis. (5 Marks)